

Deployment and Sustainability Plan

A practical plan for running VidyaSetu beyond the pilot
for first-generation, rural and low-income learners

Executive Summary

VidyaSetu is ready for a controlled pilot, not yet for uncontrolled mass rollout. The current product can onboard a learner by voice, extract a profile, generate a context-specific pathway and four-week journey, lock proof progress, surface admin visibility, and send daily WhatsApp learning reminders for approved learners. This document reports the Phase 2 deployment and sustainability plan for the production MVP, covering the current operating stack, the AI model and cost plan, the maintenance and monitoring regime, the path from pilot to district-scale adoption, and the risk controls needed for low-literacy learners, hyperlocal opportunity quality and WhatsApp outreach. The recommended next step is a 20–50 learner hyperlocal pilot run with strict consent, cost caps, worker review and partner-verified opportunities.

Keywords: deployment plan, sustainability model, voice-first onboarding, vernacular AI, proof-based journeys, Anthropic Claude, Sarvam voice, WhatsApp retention, hyperlocal pilot.

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1 Current Deployment Snapshot

VidyaSetu is deployed as a web MVP with a production URL, serverless backend functions, persistent learner data, and a working AI layer. The product is intentionally web-first so that evaluators and field partners can open it on common Android phones without installing an app. Table 1 summarises the current implementation against each layer of the stack and the sustainability decision attached to it.

Table 1. Current implementation and sustainability decision, by layer.

Layer	Current implementation	Sustainability decision
Frontend and deployment	React/Vite web app deployed on Vercel at https://vidyasetu-mvp.vercel.app	Keep Vercel for pilot speed. Move to stricter production controls only after field metrics justify scale.
Backend	Vercel serverless API routes with shared logic in <code>api/_lib/*</code> to keep the function count controlled	Avoid adding new serverless files unless absolutely required; keep shared services testable.
Database	Supabase persistence for learner profiles, chat memory, journeys, reminders and admin CRM views	Use managed Supabase for the pilot; add RLS policy review and backup routines before partner rollout.
AI planning	Anthropic Claude generates counselor reasoning, pathway options, learner journeys, and web-search-assisted resource discovery	Use a Claude-only stack for consistency, with cheaper Haiku-first generation where quality is acceptable and Sonnet fallback for robustness.
Voice and vernacular	Sarvam STT/TTS for the Indian-language speech layer, with text fallback when voice fails	Measure voice success rate and latency on low-end Android during the pilot.
WhatsApp reminders	WATI template flow configured for daily task reminders; live sending currently restricted by allowlists	Scale only after explicit learner consent, opt-out handling, and message-volume caps are visible in the product.
Admin CRM	Admin view shows learner profiles, journeys, progress, reminder state and support follow-up needs	Use the CRM as the field-worker cockpit for pilot follow-up, not just a demo dashboard.

Deployment principle. The MVP should stay narrow and reliable: two or three strong learner flows, verified resources, consent-aware reminders, and admin follow-up. The platform should not promise jobs, loans, or contacts before proof and source checks.

2 Operating Architecture

Table 2 traces each user-facing flow to the system component that serves it and the data or trigger that results.

Table 2. Operating architecture: flow, component, and stored or triggered outcome.

User flow	System component	What is stored or triggered
Voice-first intake	Meera counselor API	Learner name, goal, education, language, location, phone access, time, mobility, constraints, and profile confidence.
Pathway selection	Claude pathway planner	Three compact pathway options. The user can compare options; the pathway is locked only once a journey is created.
Learning journey	Claude journey generator	Week-by-week and day-by-day tasks, free goal-specific resources, proof requirements, and safety checks.
Proof and passport	Progress and resume/passport service	Proof notes, completion state, and a shareable Skill Passport only when proof is ready.
Daily retention	ADEWS plus WATI reminder job	Daily WhatsApp message with today’s task and resource. Future loop: silence alert and worker call after defined inactivity.
Central operations	Admin CRM	Learner status, pathway lock, proof completion, reminder outcomes, support flags, and opportunity-review needs.

2.1 Current Safety Gates

- No fabricated jobs, employers, training centres, or phone numbers.
- Offline opportunity sharing requires source/contact verification and learner consent.
- WhatsApp reminders are currently live only for approved test learners through allowlists.
- The Skill Passport should represent proof readiness, not a generic resume.
- Pathway and journey generation should use the learner’s exact goal, not a generic category label.

3 AI Model, Latency, and Cost Plan

VidyaSetu’s AI stack is Claude-led. The production configuration currently uses Anthropic model environment variables, with Sonnet used for higher-reliability counseling and Haiku-first generation used where speed and cost matter. OpenAI and Fireworks are not part of the intended primary flow. Table 3 sets out the model strategy and expected latency per task.

Table 3. Model strategy and expected latency, by task.

Task	Model strategy	Why this choice	Expected latency
Meera counselor	Claude Sonnet-class model from ANTHROPIC_MODEL	The counselor needs better instruction following, memory use, and goal stability, because profile errors break the whole product	2–8 sec typical; 20 sec timeout in the current configuration
Pathway generation	Claude Haiku-first with Sonnet fallback for difficult cases	Pathways need structured output and local realism. Haiku keeps costs lower; Sonnet fallback protects quality	8–30 sec without live web search; 20–50 sec with search
Learner journey generation	Claude Haiku-first with a longer timeout and Sonnet fallback	The journey needs detailed day-level planning, resources, and proof tasks, but can run behind a loading screen	25–90 sec depending on resources and prompt size
Resource discovery	Claude web search where enabled	Used only for specific resource discovery, not on every request	Adds search latency and cost; best used after pathway intent is clear
Voice	Sarvam STT/TTS plus text fallback	Keeps the product accessible to users who cannot type easily	1–5 sec typical per speech operation; must be field-tested on low-end Android

Pricing source and model notes. Anthropic public pricing checked on 21 June 2026: Sonnet 4.5/4.6 is listed at \$3 per million input tokens and \$15 per million output tokens; Haiku 4.5 is listed at \$1 per million input tokens and \$5 per million output tokens. Claude web search is listed at \$10 per 1,000 searches plus normal token costs. Prices can change, so procurement should re-check vendor pricing before launch.

3.1 Per-Learner AI Cost Estimate

Table 4 converts the model strategy above into a planning-level cost per learner under light, complete, and heavy usage, and at pilot and early-scale cohort sizes.

Table 4. Per-learner and per-cohort AI cost estimates.

Usage scenario	Assumption	Estimated Claude cost	Planning note
Light learner	5–6 counselor turns only; no journey regeneration	\$0.05–\$0.12, roughly Rs 4–Rs 10	Useful for profile-only users or incomplete onboarding
Complete active learner	Counselor, one pathway generation, one journey generation, and 0–2 search calls	\$0.15–\$0.35, roughly Rs 13–Rs 30	This is the main pilot budgeting number
Heavy learner	Many chat turns, repeated pathway/journey refreshes, web search, and long voice use	\$0.50–\$1.50+ per learner	Needs regeneration limits, caching, and admin review
50-learner pilot	Most learners complete one full flow, with a buffer for retries	Claude AI budget roughly \$8–\$25; all-in pilot budget should reserve \$25–\$75+	All-in includes voice, WhatsApp, infra, retries, and a monitoring buffer
1,000 active learners	One complete pathway and journey per learner	Claude AI roughly \$150–\$350; all-in may be \$300–\$900+	Actual cost depends heavily on voice minutes, WhatsApp fees, and repeat generation

The INR figures use a conservative planning conversion of about Rs 83 per USD and should be refreshed before a paid pilot. Sarvam, WATI, Vercel, Supabase, taxes, and plan fees are not included in the Claude-only estimates.

4 Maintenance and Operating Plan

Table 5 sets out the operating cadence, owner, and evidence to track for ongoing maintenance during and after the pilot.

Table 5. Maintenance cadence, ownership, and tracked evidence.

Cadence	Owner	Work to be done	Evidence to track
Daily during pilot	Field operations owner	Check the admin CRM for stuck learners, failed reminders, proof pending, silence alerts, and unsafe opportunity requests	Reminder sends, failed sends, learner return rate, proof saved, worker follow-up notes
Daily during pilot	Technical owner	Review API errors, model time-outs, malformed JSON, slow generation, and cost spikes	Vercel logs, Supabase logs, Anthropic usage, WATI send logs, Sarvam failure logs
Weekly	Product and field team	Review real conversations for repeated questions, misunderstood goals, and language-quality issues	10 sampled chats per week, profile accuracy score, learner comprehension notes
Weekly	Source and opportunity curator	Verify local centres, employers, schemes, fees, and phone/contact information before showing or sharing	Verified source sheet, date checked, worker name, source URL or visit note
Monthly	Project lead	Cost review, privacy review, template-compliance review, and partner-feedback review	Cost per active learner, support load, opt-out rate, conversion to proof and opportunity

4.1 Technical Release Gates

- `npm run build` must pass before deployment.
- The counselor test suite should pass for goal stability, exact-skill extraction, and post-profile context replies.
- Vernacular tests should include at least five languages across job, skill-training, and small-business personas.
- Slice5 and Slice6 smoke tests should remain green before mentor or judge demos.
- No secrets should be committed; keys stay only in the Vercel/project environment configuration.

4.2 Data and Consent Maintenance

The pilot must make consent explicit before sending WhatsApp messages, sharing profile details, showing proof to partners, or preparing employer outreach. Learners should be able to stop reminders, change a phone number, or say “this is not me” without losing dignity or data safety.

5 Adoption Path

Table 6 lays out the staged adoption path from mentor review through district-level scale, with the go/no-go evidence expected at each stage.

Table 6. Staged adoption path from review to district-level scale.

Stage	Timeline	Target / operating focus	Go/no-go evidence
Submission and mentor review	Now	Show two polished flows, admin CRM, reminder proof, and the evaluation report to judges, mentors and the internal team	Mentor feedback confirms the product is understandable and worth piloting
Controlled field pilot	Weeks 1–4	Run voice intake, pathway lock, journey, daily reminders, and worker follow-up for 20–50 learners through one NGO/field partner in one or two districts	70% profile completion, 50% journey start, 30% first proof saved, low safety issues
Employer and livelihood partner loop	Weeks 5–8	Show proof-based candidate readiness to 3–5 local employers or livelihood trainers plus an NGO partner before outreach	Partners say the proof and filters reduce screening effort or improve trust
Hyperlocal expansion	Weeks 9–12	Add verified centres, block-level constraints, a stronger worker CRM, and ADEWS silence alerts for 100–250 learners across selected blocks	A repeatable operating playbook and predictable cost per active learner
District or CSR-backed scale	After pilot evidence	Formalize the support team, source verification, privacy policy, consent, and reporting for NGO, CSR, skill partner, or district-level adoption	Clear funding or partner commitment, plus documented outcomes

5.1 Pilot Roles to Start With

- **Riya-type office job flow:** a Class 12 learner near Varanasi/Basti — computer basics, typing, customer service, safe day shift.
- **Small-business flow:** mushroom, poultry, food/tiffin, or tailoring — with training, cost, buyer, supplier, scheme, and loan-risk checks.
- **Short skill-to-income flow:** mobile repair, tailoring helper, plumbing helper, or local service work — with proof-first learning tasks.

Adoption principle. VidyaSetu should not spread as a generic AI chatbot. It should spread as a local operating system for learner support: vernacular intake, a practical pathway, a daily task, proof, worker follow-up, and a verified next step.

6 Sustainability Model

The product should remain free for learners during pilots. The sustainable path is partner-funded: NGOs, CSR programs, skill partners, district initiatives, and employers all benefit when learners arrive with clearer goals, proof, and consent, instead of unfiltered applications. Table 7 sets out the main cost areas and how each is controlled and scaled, and Table 8 sets out the revenue and funding options under consideration.

Table 7. Cost areas, control mechanisms, and scale plan.

Cost area	What drives cost	Control mechanism	Scale plan
Claude AI	Chat turns, pathway/journey generations, output length, and web-search calls	Haiku-first generation, Sonnet fallback, shorter prompts, caching, regeneration limits, and a cost dashboard	Budget by active learner, not by registered user
Sarvam voice	STT/TTS minutes, retries, and long spoken content	Short voice responses, stop audio on tab change, retry limits, and text fallback	Field-test low-end phones before expanding voice-heavy use
WhatsApp/WATI	Approved template sends, broadcast volume, and Meta/WATI plan charges	Consent gate, opt-out, allowlists during the pilot, one reminder per day, no spam	Move from allowlist to cohort-level sending only after templates and consent UX are stable
Vercel/Supabase	Serverless invocations, database reads/writes, storage, and logs	Keep the API compact, avoid repeated heavy web search, monitor database usage	Upgrade plans only when pilot traffic requires it
Field operations	Worker follow-up, source verification, partner onboarding, and user support	Admin CRM, standard scripts, weekly review, and partner-owned cohorts	This becomes the main real-world cost and moat as scale grows

Table 8. Revenue and funding options, with guardrails.

Option	Who pays	Why it can work	Guardrail
CSR/NGO cohort funding	CSR team, NGO, foundation, or donor	Supports measurable learner outcomes and employability/livelihood readiness	Do not charge low-income learners in early pilots
Employer hiring support	Local employers or staffing partners	Proof-filtered learners reduce screening effort and improve trust	No pay-to-apply; no sharing without learner consent
Skill partner enablement	Training centre or livelihood partner	VidyaSetu can route learners to verified, relevant training	No fabricated or paid recommendations without disclosure and verification
District/state adoption	Government or public-private program	Can integrate with existing skilling, employment, and livelihood schemes	Requires privacy, audit, grievance, and source-verification processes

7 Risk Register and Mitigation

Table 9 records the principal risks identified during deployment planning, the mitigation already present in the MVP, and the next action required before scale.

Table 9. Risk register: severity, current mitigation, and next action.

Risk	Severity	Mitigation in MVP	Next action before scale
AI gives a generic or wrong pathway	High	Exact-goal prompts, pathway/journey tests, and no job/loan guarantee language	Human review of sampled pathways and partner feedback after every cohort
Learner misunderstands a text-heavy UI	High	Voice-first counselor, vernacular UI, short route cards, and daily tasks	Test on entry-level Android with low-literacy users and observe without coaching
Fake or unsafe opportunity information	High	No fabricated jobs, contacts, or centres; source verification required	Create a verified local source database with date, worker, and evidence
WhatsApp messages without proper consent	High	Live sending restricted through allowlists and an approved template	Add visible opt-in/opt-out and a consent audit trail before wider sending
AI or voice costs rise with usage	Medium	Cost estimates, model tiering, search throttling, and generation limits	Add a per-cohort budget cap and a daily cost report
Data privacy and partner misuse	High	Consent-first profile and proof-sharing design	Add a formal privacy policy, role-based admin access, and periodic access review
Hyperlocal mismatch	Medium	Profile captures district/location and commute where available	Capture block/panchayat and verify centres/employers at that level

Responsible AI stance. VidyaSetu should be judged not only on whether AI generates content, but on whether the product prevents harmful overconfidence. The system must say when something is not verified, avoid fake opportunities, ask for missing context, and involve a worker before high-risk actions such as loans, relocation, employer outreach, or sharing learner data.

8 90-Day Execution Roadmap

Table 10 sets out the build/operate and field-validation work planned for each 90-day window, together with the decision output expected at the end of the window. Table 11 lists the pilot success metrics that will be used to assess readiness for scale.

Table 10. 90-day roadmap: build/operate, field validation, and decision output.

Window	Build/operate	Field validation	Decision output
Days 1–14	Freeze demo flows, add a consent UI for reminders, clean the admin CRM, and document the support script	Run 5–10 guided sessions with representative users and one field worker	Ready/not-ready checklist for the pilot
Days 15–35	Start a 20–50 learner cohort, run daily reminders, and track proof and support in the CRM	Observe onboarding, voice comprehension, pathway trust, and Week 1 completion	Cohort report: completion, cost, issues, and learner quotes
Days 36–60	Add verified local centres/employers/schemes for the pilot districts	Show the proof-passport to employers/NGO partners and collect feedback	Partner validation: whether proof improves trust and screening
Days 61–90	Add the ADEWS silence-alert loop, tighten privacy/access controls, and optimize model cost	Run a second cohort or an expansion in one more block/district	Scale decision with budget, staffing, and measurable outcomes

Table 11. Pilot success metrics and initial targets.

Metric	Why it matters	Initial target
Profile completion rate	Shows whether voice onboarding is understandable	70%+
Pathway trust/comprehension	Shows whether learners understand the recommended route	Average 4/5 from observed sessions
Journey start rate	Shows whether a pathway turns into action	50%+
Week 1 proof saved	Shows proof-first behavior, not only chatbot usage	30%+
Reminder delivery and return rate	Shows whether the WhatsApp loop improves retention	80%+ sent; measurable return lift
Cost per complete learner	Keeps sustainability grounded	Rs 10–Rs 30 AI cost target; all-in tracked separately
Unsafe/fabricated opportunity incidents	Protects learner trust	Zero tolerance; any incident blocks scale

9 Final Sustainability Position

VidyaSetu’s immediate sustainability path is not mass advertising or a broad consumer chatbot. The strongest path is a partner-led cohort model: one NGO or district partner brings learners, VidyaSetu provides voice-first onboarding and personalized journeys, field workers use the admin CRM to support stuck learners, and verified employers or livelihood partners receive only consent-backed proof when learners are ready.

The product can remain affordable if generation is limited to meaningful moments: counselor profile extraction, one pathway set, one four-week journey, and daily short reminders. The largest long-term cost is not only AI tokens; it is the human operating layer required to verify

local sources, support learners, and keep opportunities honest. That operating layer is also what differentiates VidyaSetu from a generic chatbot.

For Phase 2, the recommended claim is that VidyaSetu is a functional, AI-backed MVP ready for a controlled hyperlocal pilot. It has a live deployment, a Claude-based AI engine, vernacular voice-first onboarding, pathway and journey generation, proof/passport gating, admin CRM visibility, and WhatsApp reminder support. It should now be tested with 20–50 learners and one field partner before scaling.

9.1 Final Assessment

VidyaSetu is ready as an MVP for a focused, hyperlocal pilot. It demonstrates real AI functionality, vernacular voice-first onboarding, proof-based journey creation, responsible opportunity handling, admin visibility and WhatsApp reminder support. The product is not yet a scaled field deployment, but it has enough functional and operating depth to justify pilot testing with NGO and employer partners.

One-line adoption ask. Partner with VidyaSetu for a 20–50 learner pilot in one district: voice onboarding, a personalized pathway, daily proof tasks, WhatsApp reminders, worker follow-up, and verified next-step matching.

9.2 References Used for Cost Assumptions

- Anthropic public pricing page: <https://www.anthropic.com/pricing>
- Anthropic Claude API pricing docs: <https://platform.claude.com/docs/en/about-claude/pricing>
- Current VidyaSetu production configuration and implemented feature set as of 21 June 2026.